CNC Procedure 2016_030	Sep 26th 2016
Procedure for αCZi sensor Assembly	Can be used as reference guide for other sensor model also

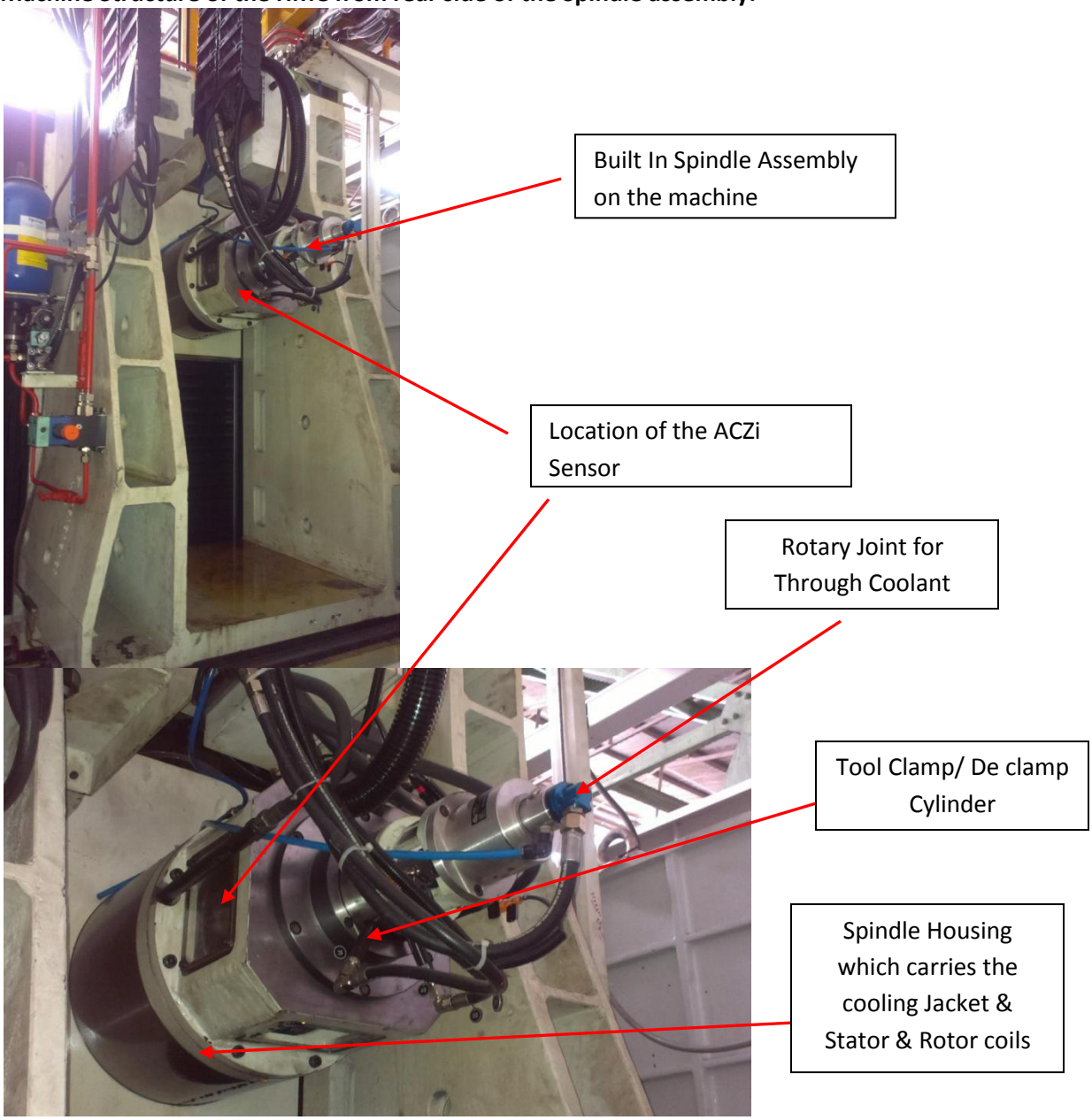
Applicable Machine Type: HMC**/VMC/VTL with built-in spindle units**

Spindle Type: Built In Spindle with FANUC Built In Spindle Motor

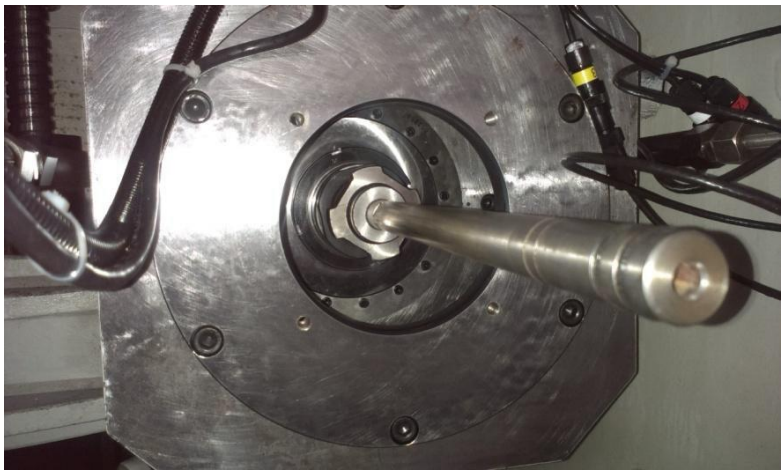
<u>The below mentioned sensor is shown in the photographs below</u>:

αCZi Sensor: 1024 I (A860-2161-T611),
No of teeth – 1024, Max Speed – 8000 RPM

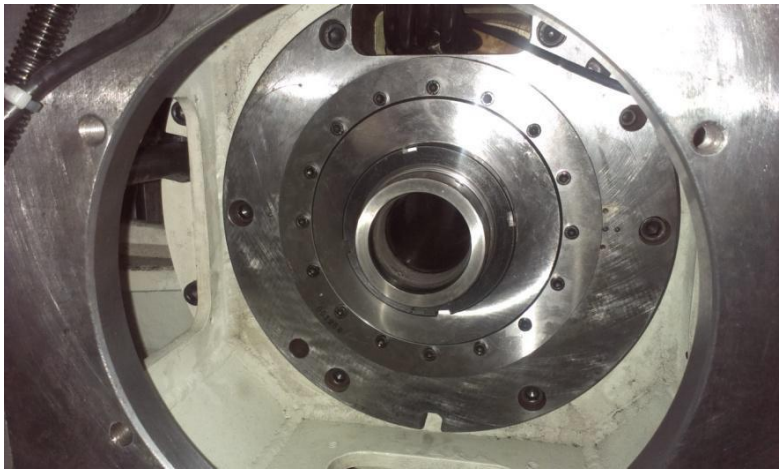
Machine Structure of the HMC from rear side of the Spindle assembly:



The αCZi sensor assembly is placed below the clamp/ declamps cylinder and the rotary joint. Hence it is necessary to remove the assembly so that clear access is available for assembling the sensor head. Always insist on removing the subassemblies before checking the αCZi sensor.



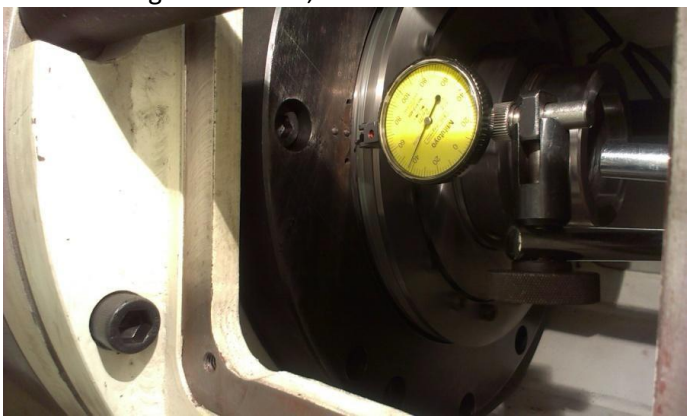
Spindle rear side showing the Coolant rotary joint and the clamp/ declamp cylinder removed and the drawbar still in place.



Spindle rear side showing the Coolant rotary joint and the clamp/ declamp cylinder and the drawbar removed.

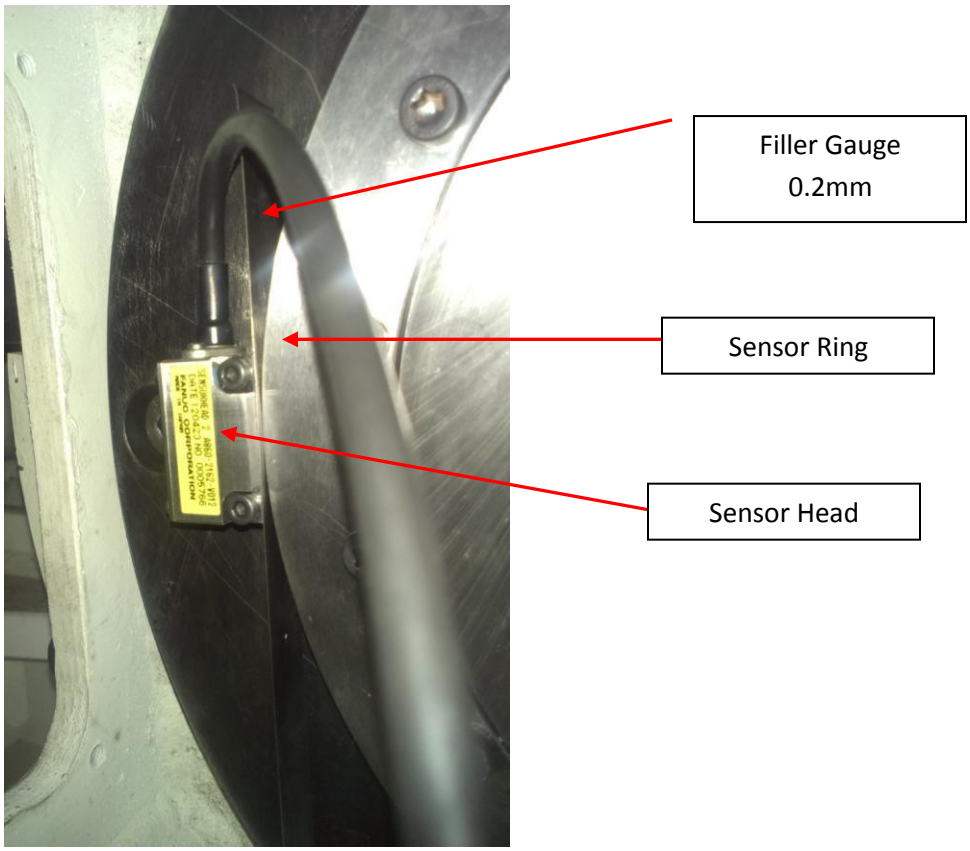
αCZi sensor assembly to be tested in this condition.

Once the ACZi sensor assembly is exposed, we should first ensure the run out on the CZi sensor ring with the help of 1 µm dial gauge. The run out is supposed to be within 10µm, practically we have measured the run out on this machine to be within 27 µm, as the mechanical construction was such that there was no provision for correcting this run out, we retained the run out to be within 27 µm.

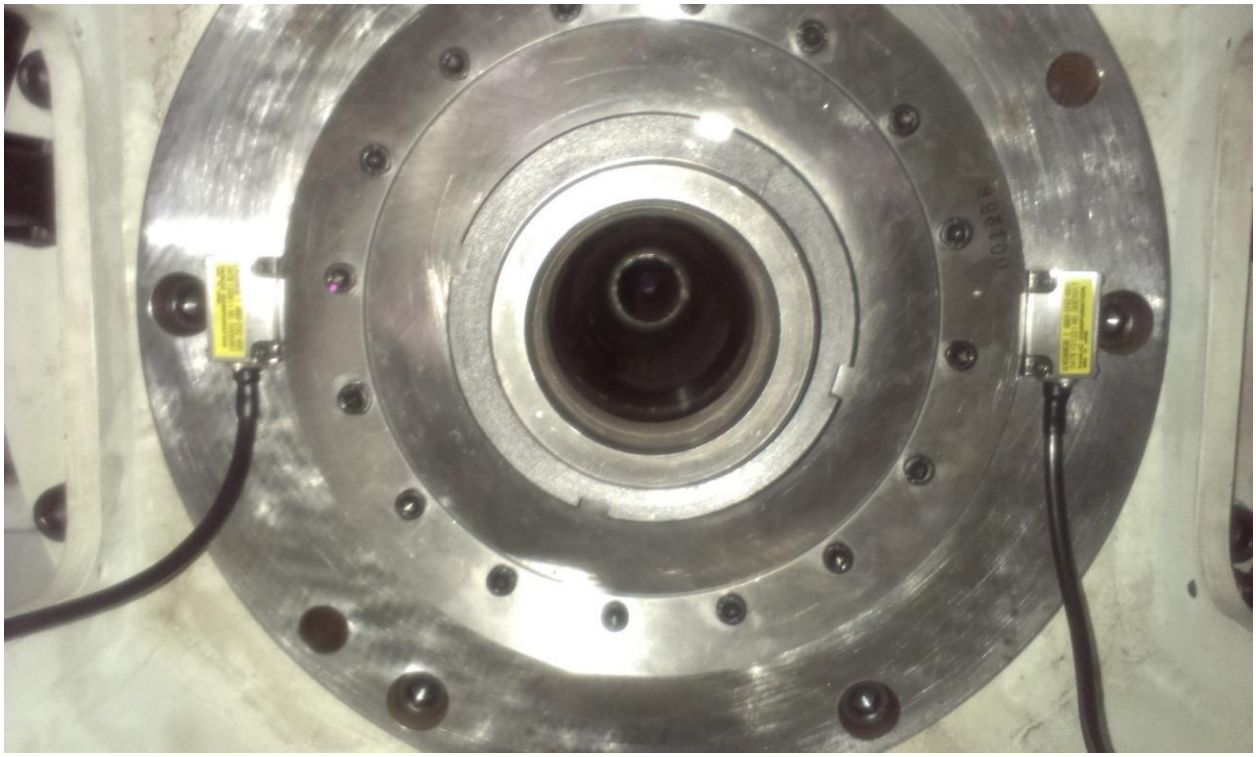


The sensor head assembly is then done, the sensors are fixed to the sensor head mounting plate with the help of the dowels and the thread locker. Thread locker – ***Loctite 242*** can be used. Thread locker is applied as shown in the photograph below.

The head is then mounted with the help of the filler gauge of 0.2 mm. In the manual it is recommended to maintain a gap of 0.1 mm however as the run out on the ring was 27µm, we increased the spacing from 0.1 mm to 0.2mm

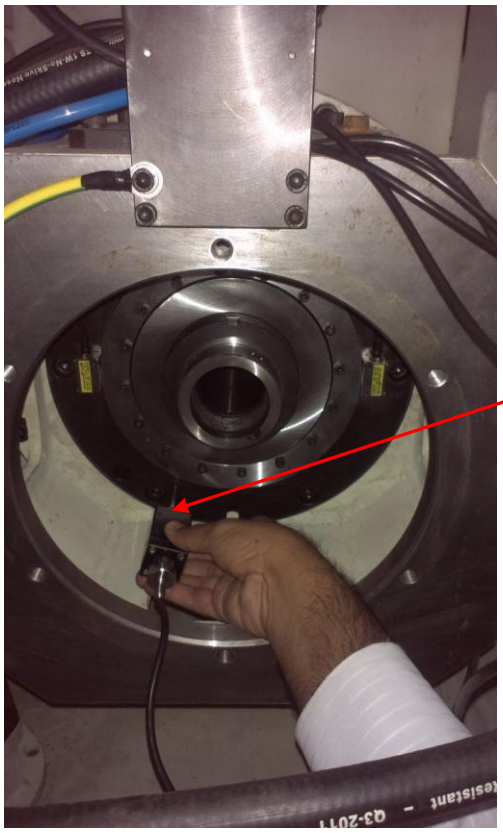


Both the sensor heads are then mounted and the assembly is completed as below:



The spindle is then rotated upto maximum speed (6000 RPM in this case), in steps of 500 RPM. Vibration sensor is used to record the vibration in the axial and the radial direction on the sensor head mounting plate.

Vibration Meter used for measuring vibrations.

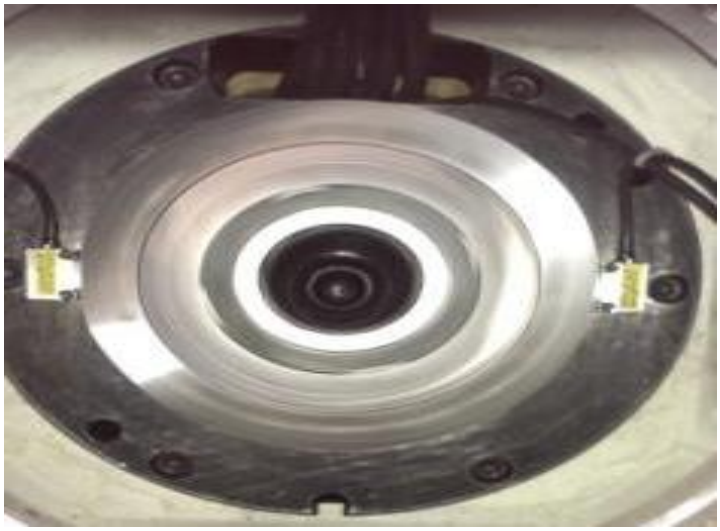


Vibration Sensor used for measuring vibrations in the axial direction, on the sensor head mounting surface.

Some reference reading of the vibration in axial & radial directions: The vibration limit specified is 1G in the manual; the vibration was well within the specified limit.

Sr. No.	Spindle Speed (RPM)	Vibration Level (G) (Axial Direction)	Vibration Level (G) (Radial Direction)
1	500	0.01	0.01
2	1000	0.01	0.01
3	1500	0.03	0.03
4	2000	0.02	0.02
5	2500	0.02	0.02
6	3000	0.03	0.03
7	3500	0.02	0.03
8	4000	0.03	0.03
9	4500	0.04	0.05
10	5000	0.09	0.08
11	5500	0.1	0.1
12	6000	0.1	0.1

The spindle assembly has to be completed, with the drawbar, the clamping mechanism and the rotary joint for the coolant. The vibrations are again measured with the assembly, reference readings are as mentioned below:

Sr. No.	Spindle Speed (RPM)	Vibration Level (G) (Axial Direction)	Vibration Level (G) (Radial Direction)
1	500	0.03	0.02
2	1000	0.04	0.03
3	1500	0.05	0.04
4	2000	0.05	0.05
5	2500	0.1	0.06
6	3000	0.25	0.15
7	3500	0.25	0.15
8	4000	0.35	0.15
9	4500	0.45	0.25
10	5000	0.5	0.25
11	5500	0.6	0.2
12	6000	0.6	0.2

The αCZi sensor assembly is complete.